

Question Paper

Exam Date & Time: 18-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Digital design verifications [ECE_4407]

Marks: 50

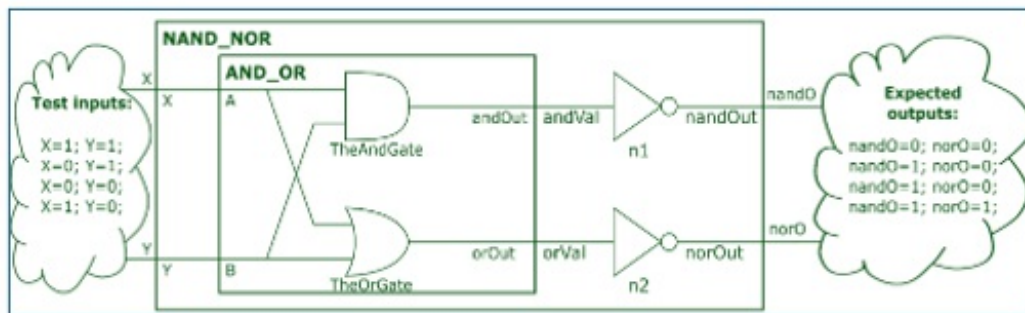
Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Write the test bench program of NAND-NOR logic in System Verilog with set of Test inputs and expected outputs as shown in the Figure below. (5)

A)



Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Explain the working of a Full Adder using its truth table and logic diagram and write the corresponding Verilog code. (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- C) Explain the importance of a Verification Plan (V-Plan). (2)

Course Outcome: "CO1" / Cognitive Level: "Remember"

- 2) Analyze different types of verification, its goals, and various verification methodologies, explaining how they interrelate to ensure a reliable VLSI design. (5)

A)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Compute and contrast between Verilog, VHDL and System Verilog with a programming example of HALF adder. (3)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- C) Explain any two advantages of logic verification in the digital design process. (2)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 3) Discuss the challenges in logic verification at lower technology nodes (e.g., 5 nm) with example and suggest strategies to overcome them. (5)

- A) Course Outcome: "CO4" / Cognitive Level: "Application"
- B) Apply the concept of functional coverage to a System Verilog verification environment and explain the way it measures the completeness of test scenarios in a design verification process. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Design a simple testbench for an AND gate and explain its components. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Apply UVM and explain block diagram of a UVM environment to verify an ALU (Arithmetic Logic Unit). (5)

- A) Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Apply the concept of digital design verification to explain how it ensures correctness and reliability within the VLSI design flow, using suitable examples from practical design scenarios. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Explain the role of Universal Verification Methodology (UVM) in modern verification environments and how it improves the verification process. (2)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5) Analyze how a race condition can occur in a simulation testbench and describe the methods and tools you would use to detect and debug it. (5)

- A) Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Discuss the steps to compile, elaborate, and simulate a design in Verilog. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) Explain the need for Synthesis in VLSI design. (2)

Course Outcome: "CO1" / Cognitive Level: "Understand"

-----End-----

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 ECE_4407:Digital design verifications

#	Course Outcome	Description	Marks
1	1	CO 1	12
2	2	CO 2	10
3	3	CO 3	5
4	4	CO 4	12
5	5	CO 5	11
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 ECE_4407:Digital design verifications

	No of	
--	-------	--

#	Level	Questions	Marks (%)
1	Remember	1	2(4%)
2	Understand	4	9(18%)
3	Application	6	23(46%)
4	Analysis	4	16(32%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)